

第四次作业

4.7

employee (ID, person_name, street, city)
works (ID, company_name, salary)
company (company_name, city)
manages (ID, manager_id)

Figure 4.12 Employee database.

Consider the employee database of Figure 4.12. Give an SQL DDL definition of this database. Identify referential-integrity constraints that should hold, and include them in the DDL definition.

```
CREATE TABLE employee (  
    id INTEGER,  
    person_name VARCHAR(50),  
    street VARCHAR(50),  
    city VARCHAR(50),  
    PRIMARY KEY (id)  
);  
  
CREATE TABLE company (  
    company_name VARCHAR(50),  
    city VARCHAR(50),  
    PRIMARY KEY(company_name)  
);  
  
CREATE TABLE works (  
    id INTEGER,  
    company_name VARCHAR(50),  
    salary numeric(10,2),  
    PRIMARY KEY(id),  
    FOREIGN KEY (id) REFERENCES employee(id),  
    FOREIGN KEY (company_name) REFERENCES company(company_name)  
);  
  
CREATE TABLE manages (  
    id INTEGER,  
    manager_id INTEGER,  
    PRIMARY KEY (id),  
    FOREIGN KEY (id) REFERENCES employee (id),  
    FOREIGN KEY (manager_id) REFERENCES employee (id)  
)
```

4.18

For the database of Figure 4.12, write a query to find the ID of each employee with no manager. Note that an employee may simply have no manager listed or may have a *null* manager. Write your query using an outer join and then write it again using no outer join at all.

- Using an outer join:

```
SELECT e.id
FROM employee e LEFT OUTER JOIN manages m
  ON e.id = m.id
WHERE m.id IS NULL
  OR m.manager_id IS NULL;
```

- Using no outer join:

```
SELECT id
FROM employee
WHERE id NOT IN (
  SELECT id
  FROM manages
  WHERE manager_id IS NOT NULL
);
```

5.4

Describe the circumstances in which you would choose to use embedded SQL rather than SQL alone or only a general-purpose programming language.

通常来说，用 SQL 编写查询语句，比用通用编程语言去实现同样的查询要容易得多。不过，并不是所有类型的查询都能用 SQL 表达出来。另外，像打印报表、与用户交互，或者把查询结果发送到图形界面这类非声明式操作，也无法单靠 SQL 在内部完成。

因此，在一些既希望利用 SQL 的简洁高效、又希望保留通用编程语言灵活性的场景下，我们就可以选择使用嵌入式 SQL 或动态 SQL，而不是只使用 SQL，或者只使用通用编程语言。

5.15

Consider an employee database with two relations:

employee (employee_name, street, city)
works (employee_name, company_name, salary)

where the primary keys are underlined. Write a function *avg_salary* that takes a company name as an argument and finds the average salary of employees at that company. Then, write an SQL statement, using that function, to find companies whose employees earn a higher salary, on average, than the average salary at "First Bank."

```

CREATE FUNCTION avg_salary(company_name VARCHAR(20))
RETURNS REAL
BEGIN
DECLARE retval REAL;
    SELECT AVG(salary)
    FROM works
    WHERE works.company_name = company_name;
RETURN retval;
END;

SELECT DISTINCT company_name
FROM works
WHERE avg_salary(company_name) > avg_salary('First Bank');

```

5.19

Suppose there are two relations r and s , such that the foreign key B of r references the primary key A of s . Describe how the trigger mechanism can be used to implement the **on delete cascade** option when a tuple is deleted from s .

当关系 s 中的任意一行被删除时，触发器机制会执行：删除关系 r 中所有引用了该被删除行的记录。